Jun Gao

Education

University of Ottawa - Master of Electrical and Computer Engineering - Co-op (GPA: 9.6/10) **Sep 2024 - Present**

Dalhousie University - Bachelor of Computer Science, Honours in Computer Science (GPA: 4.23/4.3) **Sep 2021 – Aug 2024**

Work Experience

AI Software Developer Co-op **Jan 2026 - Present**
Ciena ***Ottawa,ON***

- Develop Agentic AI systems to streamline internal workflows, supporting multiple teams across the company and enhancing operational efficiency.
- Develop reusable Model Context Protocol (MCP) servers to establish a standardized framework for tool and data access, enabling AI agents to interact seamlessly with internal data systems and streamlining the integration of new capabilities across multiple engineering teams.
- Architected and implemented sophisticated database integrations utilizing Retrieval-Augmented Generation (RAG) architectures to facilitate complex semantic searching and enable high-fidelity semantic context, streamlining data retrieval.
- Optimized AI agent performance through advanced context engineering, achieving a 2x reduction in response latency and a 10x decrease in token costs.

Data Analyst Co-op **Sep 2025 -Dec 2025**
Engineering Services, University of Ottawa ***Ottawa,ON***

- Analyzed time-series data from building systems to drive energy management initiatives, help 10% reduction in overall energy consumption.
- Develop and maintain dashboards using Grafana to visualize key performance indicators for engineering teams, reducing manual reporting time and increasing data visibility for 5+ engineering teams.
- Architected real-time sensor data pipelines in InfluxDB, utilizing optimized Flux queries to improve data processing speed by 20%.
- Collaborate with electrical/mechanical engineering experts to identify opportunities for efficiency improvements.

Teaching Assistant for Intelligent Health Informatics - Part Time **May 2025 – Dec 2025**
University of Ottawa ***Ottawa, ON***

- Mentor student teams in digital health app development, providing technical guidance on software architecture, data flow, and healthcare standards.
- Support students through course concepts, assess student progress and provide feedback.

Full Stack Software Developer Intern **May 2025 - Aug 2025**
CaIoT Inc. ***Remote***

- Developed responsive and dynamic frontend using Next.js to boost customer engagement.
- Implemented and maintained Python microservices backend for core features and RESTful API endpoints.
- Enhanced the AI agent system by integrating with Retrieval-Augmented Generation (RAG) and Feedback Loop, improving the success rate by 45%.
- Designed a distributed task execution system with Celery and Redis to improve scalability and reliability, achieving 99.9% uptime for background jobs.

Applied Projects

E-Hospital Platform
University of Ottawa

- Support the continued development and operations of **e-Hospital.ca**, a digital health platform led by the University of Ottawa.
- Manage the platform's AWS cloud infrastructure, established a CI/CD pipeline to automate the build, testing, and deployment onto the AWS.
- Maintain the FastAPI backend services and the React frontend, ensuring high availability and responsiveness for the application.

Voice-Enabled Human-Robot Interaction System on Unitree G1 Robot (Demo Video)
University of Ottawa

- Designed and implemented a C++ application that integrates the Unitree G1 robot SDK, enabling real-time speech interaction for custom chatbots.
- Developed a Retrieval-Augmented Generation (RAG) AI agent to provide domain-specific responses for the robot.

Skills

Languages: Python, JavaScript, Java, C/C++, HTML, CSS, SQL
Frameworks & Libraries: Next.js, React, Node.js, Flask, FastAPI, Django, TensorFlow, NumPy, Pandas, PyTorch, Scikit-learn, Celery, Redis
Tools & Platforms: MySQL, PostgreSQL, MongoDB, InfluxDB, Git, GitHub Actions, Docker, Kubernetes, AWS, Microsoft Azure, Grafana, Power BI, Linux, Shell Scripting